MET CS 777 · BIG DATA ANALYTICS · TERM PROJECT

Can the news tell us where violence is coming?

Forecasting spikes in violent conflict for 118 countries, one to four weeks ahead, from a free feed of the world's news.

Deven Shah · Boston University MET · August 2026

THE PROBLEM

Decisions get made before violence peaks, not after

1. **Aid agencies** have to pre-position supplies somewhere, weeks in advance.
2. **Peacekeeping missions** have to decide where patrols go next month.
3. **Newsrooms and analysts** have to pick which country gets attention this week.

So the question is not "which countries are dangerous?" It is "which countries are about to get worse than they usually are?"

The tools that exist move too slowly

Standard conflict-risk indices refresh **quarterly or once a year**, and they are built from slow indicators: GDP per capita, regime type, past wars.

They explain why a country is fragile. They cannot tell you which week the fragility turns into events.

THE DATA

GDELT: a machine that reads the world's news, all day

A public project scans news coverage worldwide and turns every reported incident (a protest, a raid, a border clash, a peace talk) into one row of a table: who did what to whom, roughly where, and when.

15 min
A new batch of events, around the clock

65+
Languages of news coverage read

9.1M
Events used, after cleaning

The catch, stated up front

GDELT counts news reports, not verified incidents.

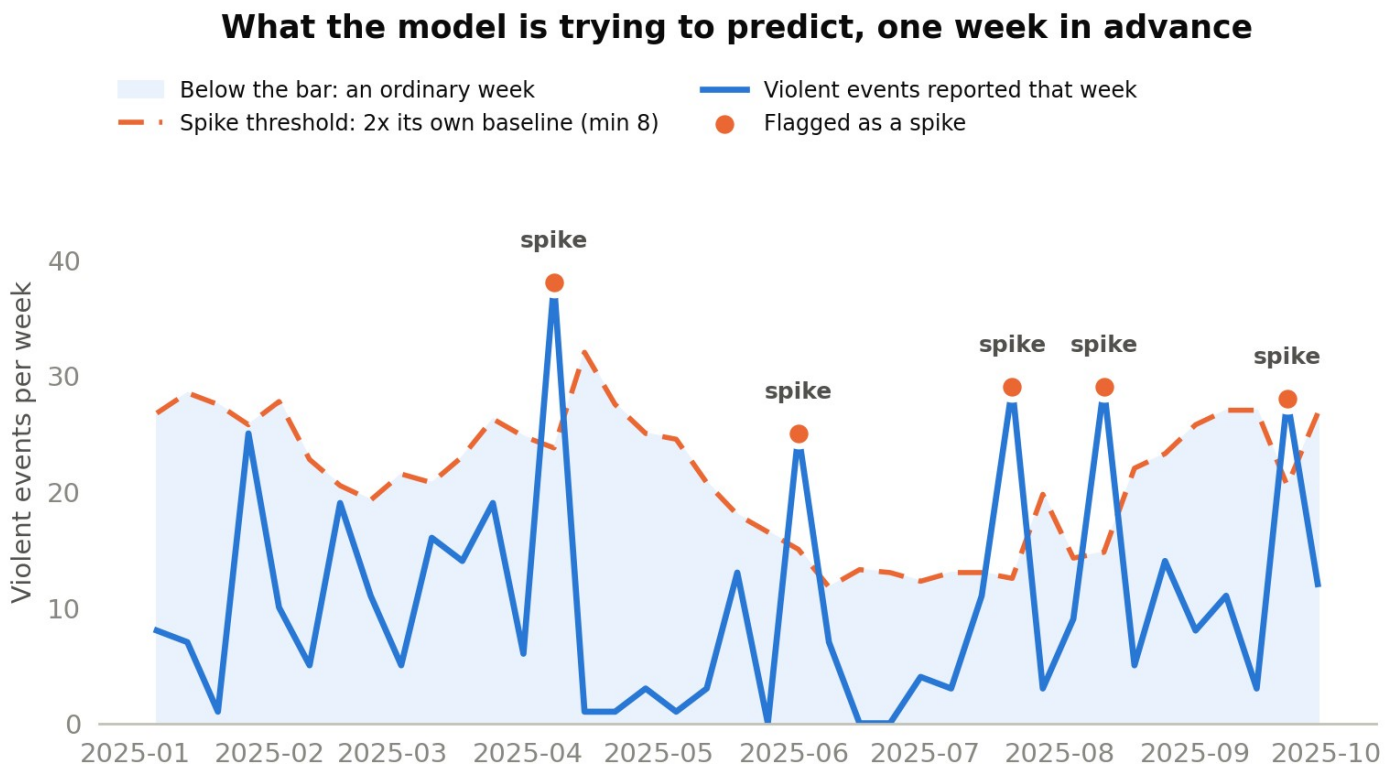
So it measures reporting about the world rather than the world itself. Coverage rises when a country becomes newsworthy for any reason: an election, a summit, a visiting correspondent.

This comes back in the limitations. It is the biggest caveat on everything that follows.

Free to download · 96 files per day · 61 columns per event · February 2015 to today

THE TARGET

"A spike" means unusual for that country



Why relative, not absolute

Absolute counts are dominated by a handful of chronically violent countries. A model trained on those would just learn a fixed ranking and look impressive while telling nobody anything new.

Escalation is the useful question.

Flagged when the violent-event count is at least twice its own trailing eight-week average, and at least eight events. The floor stops small numbers from faking a crisis.

WHY THIS IS A BIG-DATA PROBLEM

Huge at the front, small at the back

7,472
raw files downloaded

9.1M
events scanned in Spark

15,930
country-week rows out

42
features per row

12 minutes

to scan and aggregate the raw events in Spark. This is the only stage that genuinely needs a cluster to grow.

59 seconds

to train and evaluate every model at all four horizons, because the table is sized by the calendar, not by the feed.

Sizing a cluster for this workload means sizing it for one distributed scan and nothing else.

HOW IT WORKS

Five stages, one command



Download

Keeps 11 of 61 columns and resumes where it stopped.

Aggregate

Events become one row per country per week, in Spark.

Label

Flag the weeks that jumped past 2x their own baseline.

Train

Boosted trees plus two baselines, four horizons.

Serve

Score the latest week; a web app shows the board.

Everything runs on a laptop with a local Spark session. The same code scales by adding executors to the aggregation step.

MAKING IT A REAL FORECAST

Three rules that stop the model cheating

1

No peeking forward

Every rolling average and lag looks strictly backwards. A row for this week contains nothing that was not knowable by the end of this week.

Enforced in the window definition, not by memory.

2

Train on the past, test on the future

The first 75% of weeks train; the rest are the test. A gap sits between them so no training label reaches across the boundary.

A random split would let the model interpolate instead of forecast.

3

Never report accuracy

Only 9% of country-weeks are spikes, so predicting "nothing will happen" every time is 91% accurate and completely useless.

The base rate is the number every result gets compared against.

These are the choices that decide whether the numbers on the next slides mean anything at all.

THE MODELS

One candidate, two baselines chosen to be hard

Gradient-Boosted Trees

The candidate. Hundreds of small decision trees, each correcting the last. Good at messy tabular data on wildly different scales, and it reports which features it used.

80 trees · depth 4 · Spark MLlib

Logistic regression

Tests whether the complexity is earning its keep. If a straight-line model matches the ensemble, the ensemble is not adding much.

standardised features · L2

"Assume next week looks like this week"

The rule of thumb, with no parameters and no training. Conflict is bursty, so this is genuinely hard to beat, and beating it is the minimum evidence of foresight.

the real floor

All three are scored on exactly the same held-out weeks.

HOW TO KEEP SCORE

Only 9 weeks in 100 are spikes.

That single fact decides which numbers are worth reporting.

0.090

What pure guessing scores. Every result gets compared to this line.

PR-AUC

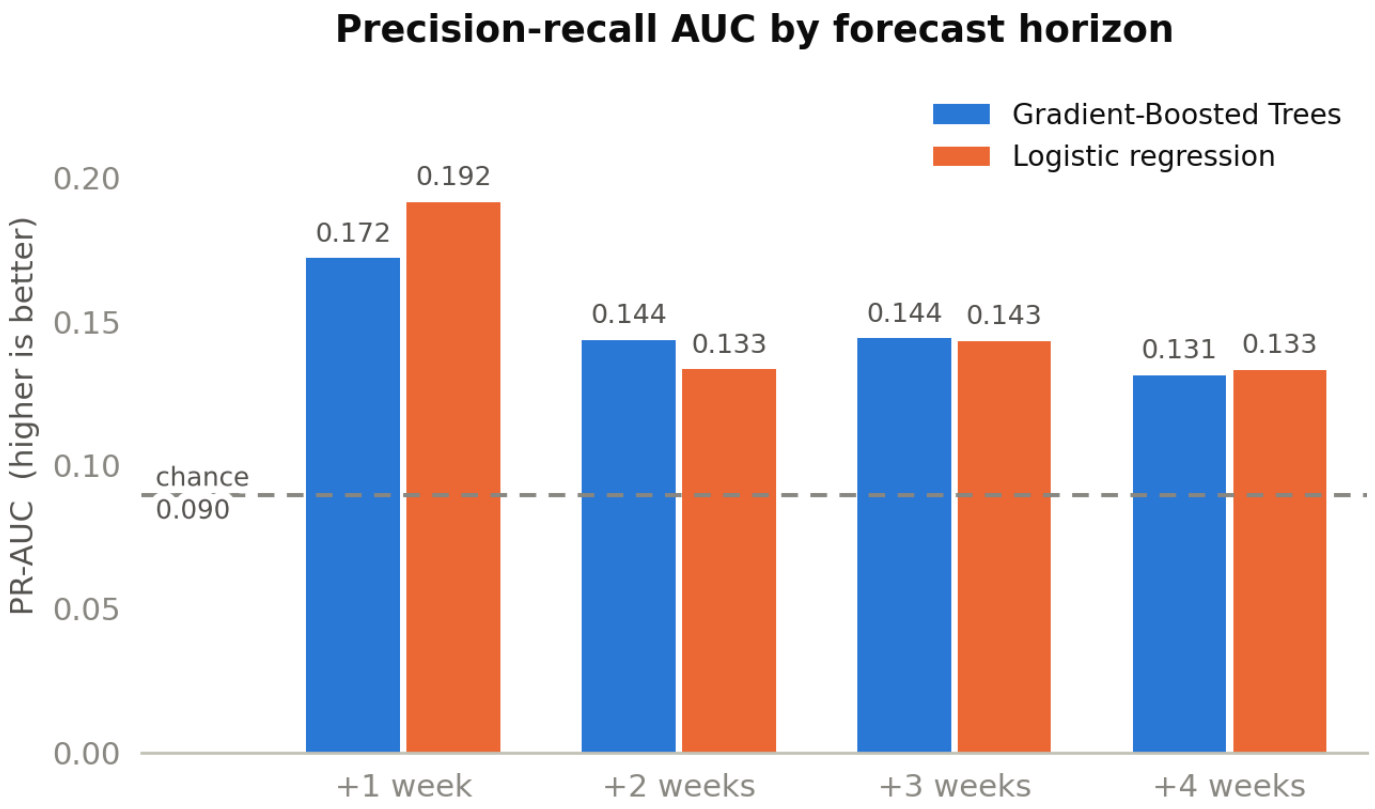
One number for the whole trade-off between catching spikes and crying wolf.

1 in 10

The alert budget. If only one quiet week in ten may raise an alarm, how many real spikes do we catch?

RESULT 1

There is a real signal, and it fades with distance



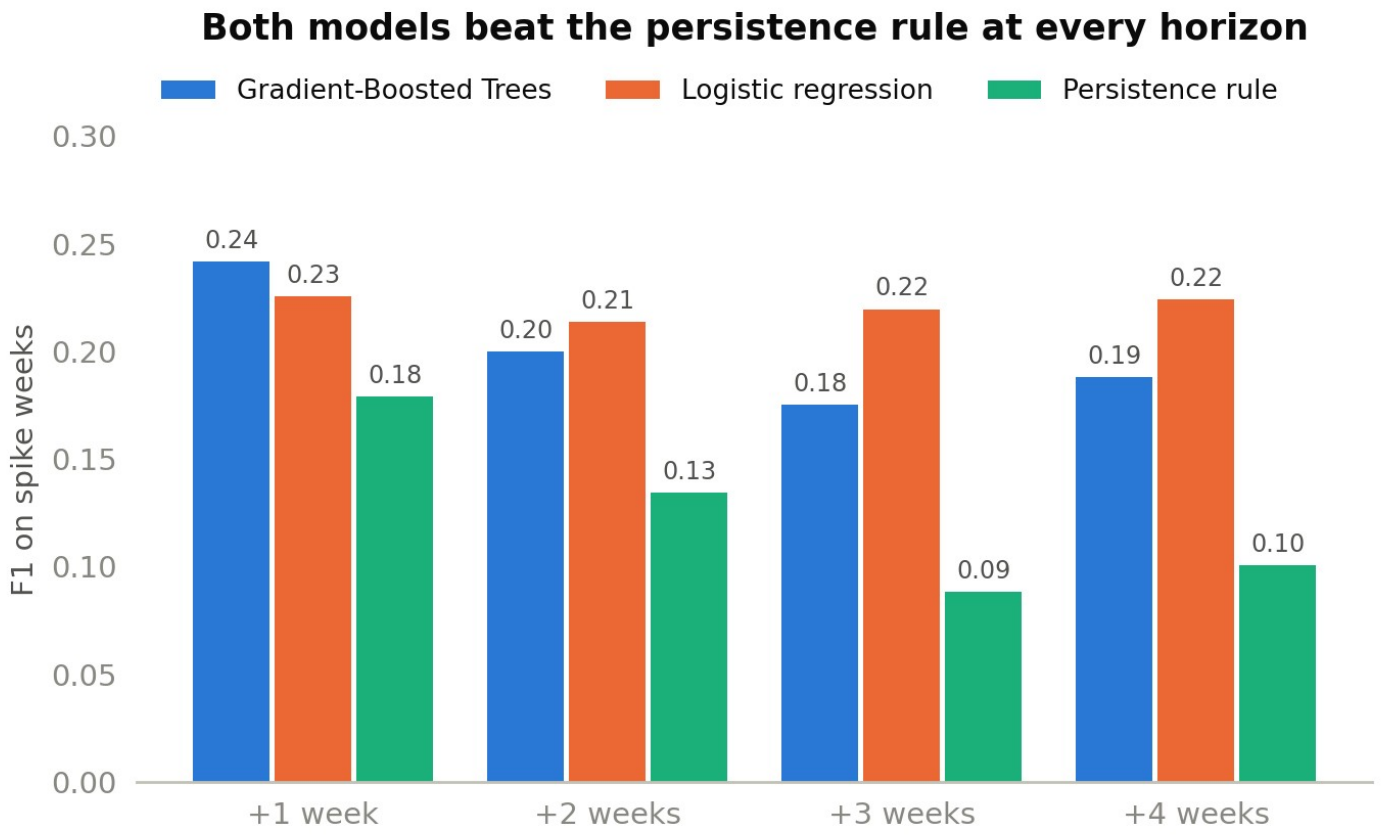
Roughly twice as good as guessing

- 0.172 one week out, against a 0.090 floor: a 1.9x lift.
- Decays smoothly to 0.131 by four weeks.
- So the useful horizon is one to two weeks, not a month.
- Straight-line model edges it at +1 week. More on that shortly.

PR-AUC summarises the whole precision/recall trade-off in one number. Chance level equals the share of weeks that are spikes.

RESULT 2

It beats the rule of thumb at every horizon



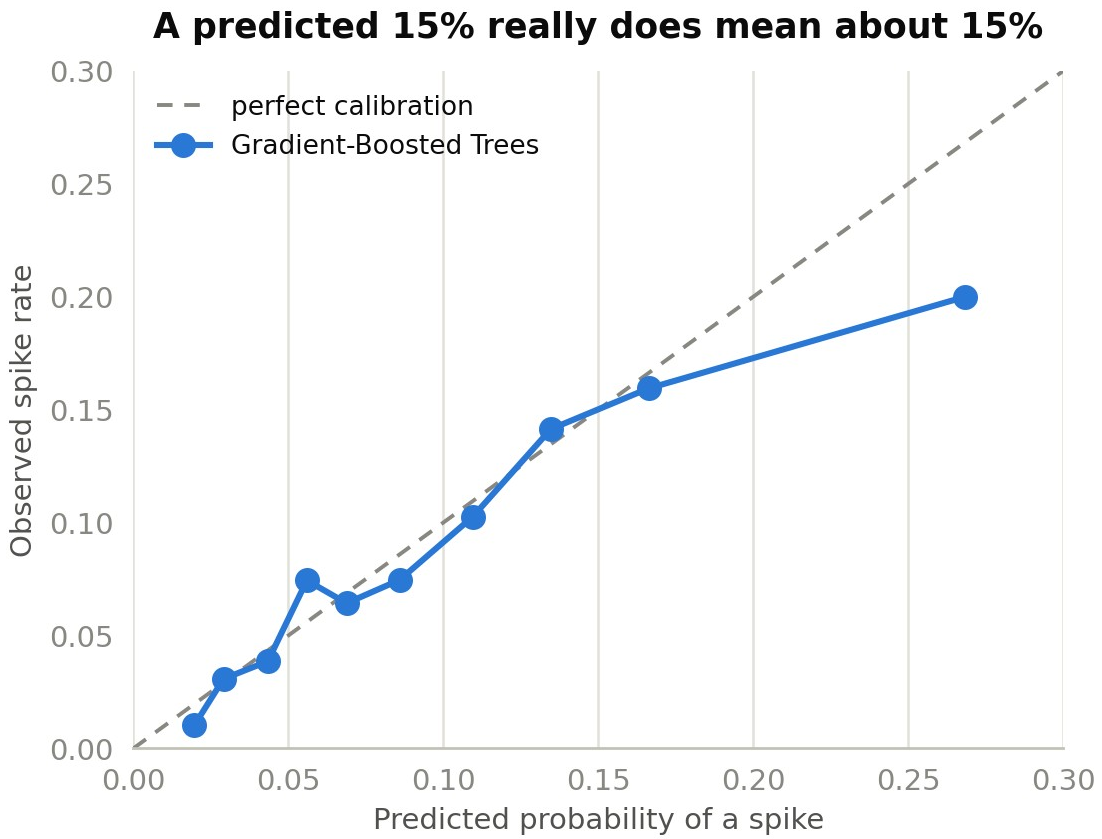
The most trustworthy finding here

- Better than "assume next week looks like this week" at every horizon.
- The margin widens: 35% better at one week, 99% at three.
- The rule of thumb collapses with distance; the model does not.
- That gap is the evidence it learned about build-up, not just inertia.

F1 balances catching real spikes against crying wolf. Higher is better; the persistence rule is the no-model floor.

RESULT 3

When it says 15%, it means about 15%



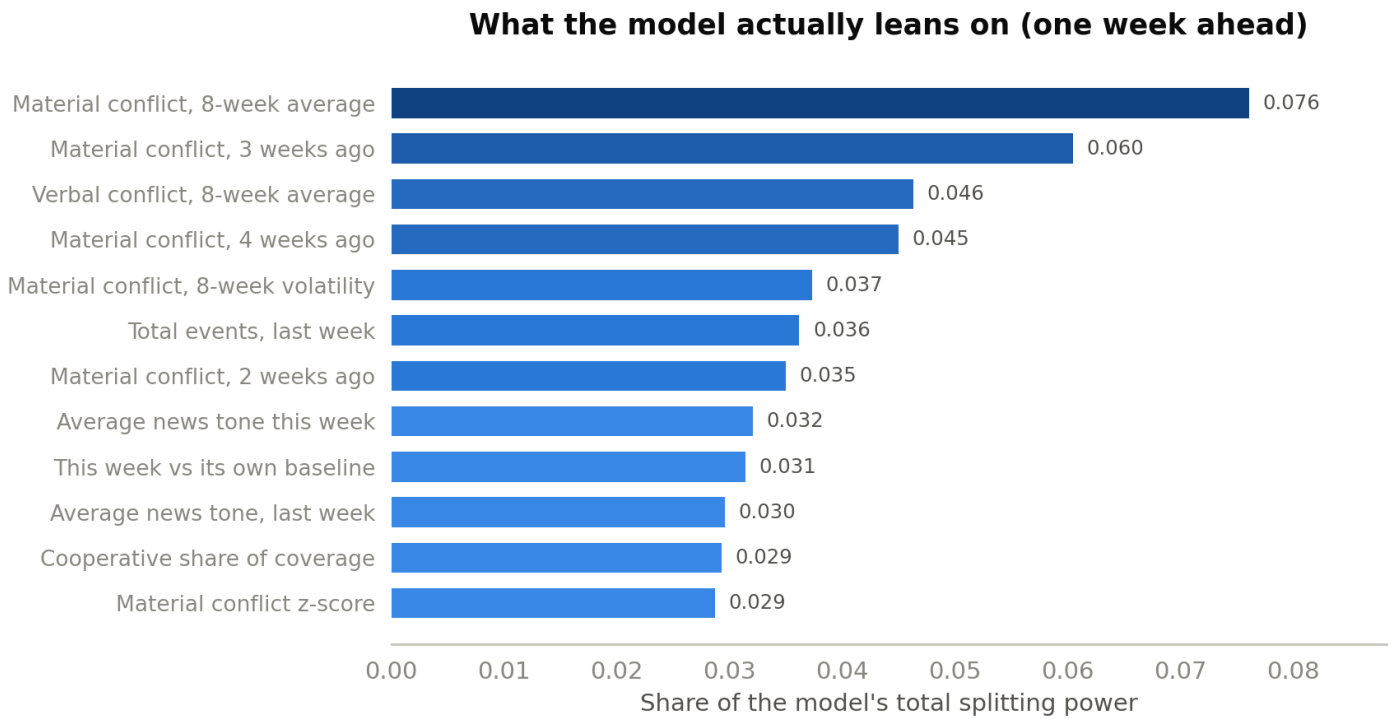
The probabilities are honest

- Across nine of ten confidence bands, predicted and observed rates track closely.
- The most confident band overshoots: 0.268 predicted against 0.200 observed.
- Nothing in the training optimises for this. It came out clean anyway.
- It matters because a calibrated score can be compared across countries and thresholded to your own risk appetite.

Each point is one tenth of the test weeks, grouped by how confident the model was. On the dashed line, the forecast is exactly right on average.

RESULT 4

Hostile words help, but less than hoped



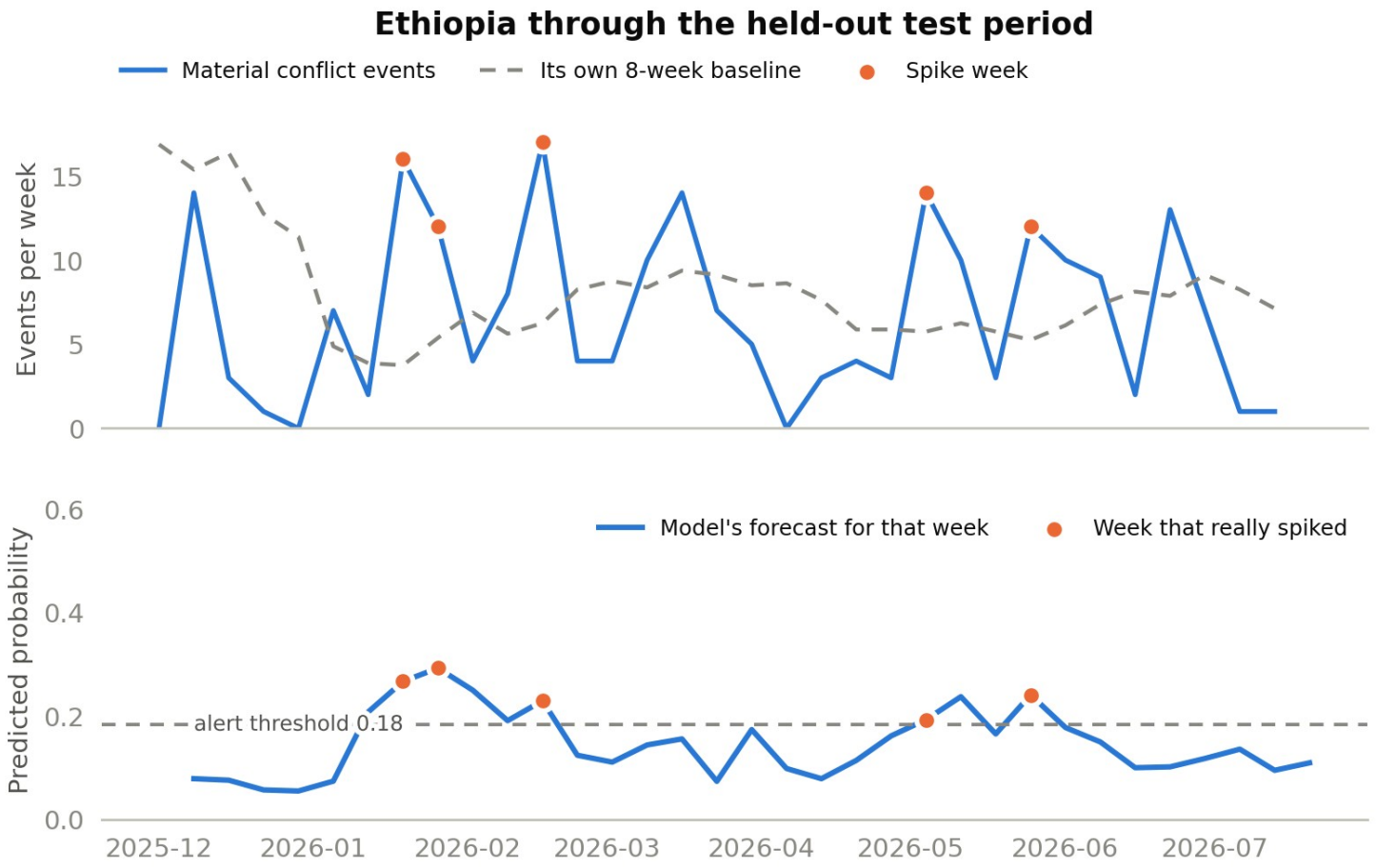
The escalation ladder, partly confirmed

- The hypothesis: threats and protests come before violence.
- Verbal conflict does rank third: real, and it survives at every horizon.
- But a country's own recent violence dominates the top of the list.
- Honest verdict: most of what is predictable here is momentum, with verbal friction adding a genuine second increment.

Importance is how much each feature contributed to the model's decisions. No single feature carries more than 7.6 percent.

ONE COUNTRY, UP CLOSE

Ethiopia through the weeks the model had never seen



What it looks like when it works

- Top: violent events each week against the country's own baseline; dots are real spikes.
- Bottom: the forecast made a week earlier for that same week.
- The forecast rises into the winter cluster, falls through the quiet spring, lifts again before May and June.
- Ethiopia is the best case in the test set, not a typical one. The spread slide is the honest counterweight.

LIVE DEMO

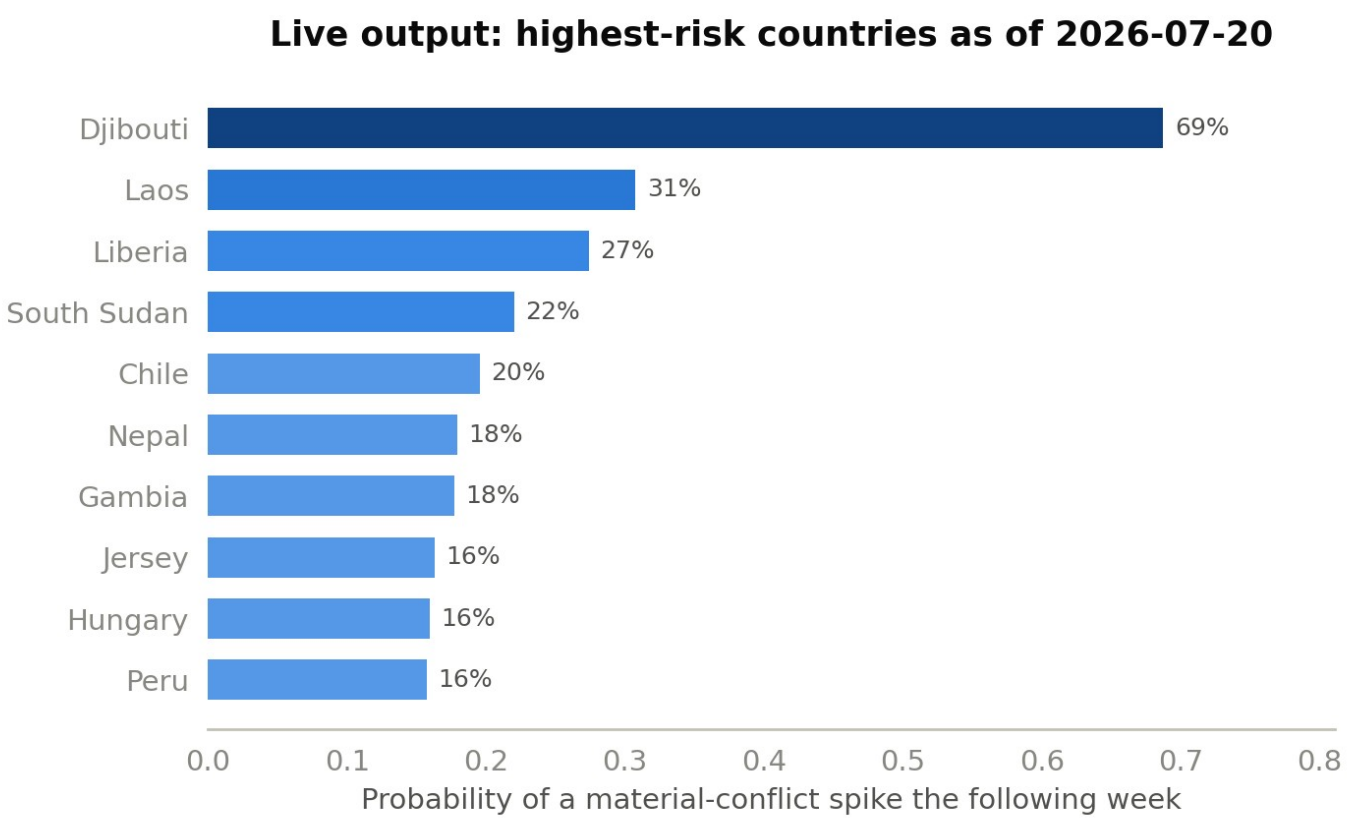
The pipeline, running

What the recording walks through

1. The risk board: all 118 countries, ranked, four horizons each
2. A single country's history with its spike weeks marked
3. The evaluation page: PR-AUC, calibration, feature importances
4. Hitting refresh: download, rebuild, retrain, rescore, live in the console

WHAT THE DEMO SHOWS

The output: every country, scored every week



As of the week of 20 July 2026

- Highest one-week risk: Djibouti at 0.69, then South Sudan at 0.33 and Laos at 0.31.
- Djibouti is instructive and a bit deflating. It was already spiking, so the model largely re-expressed the present.
- That overlap with the rule of thumb is strongest at the one-week horizon, which is exactly what the feature importances predicted.

Every country is rescored whenever the pipeline refreshes; the board below is what the web application renders.

WHAT IT CAN AND CANNOT DO

The same result, translated into alerts

Over the 33 test weeks, at an alert budget of one false alarm per ten quiet country-weeks:

442	87	349	1 in 5
alerts raised	were real spikes	spikes actually happened	alerts were right

Good for

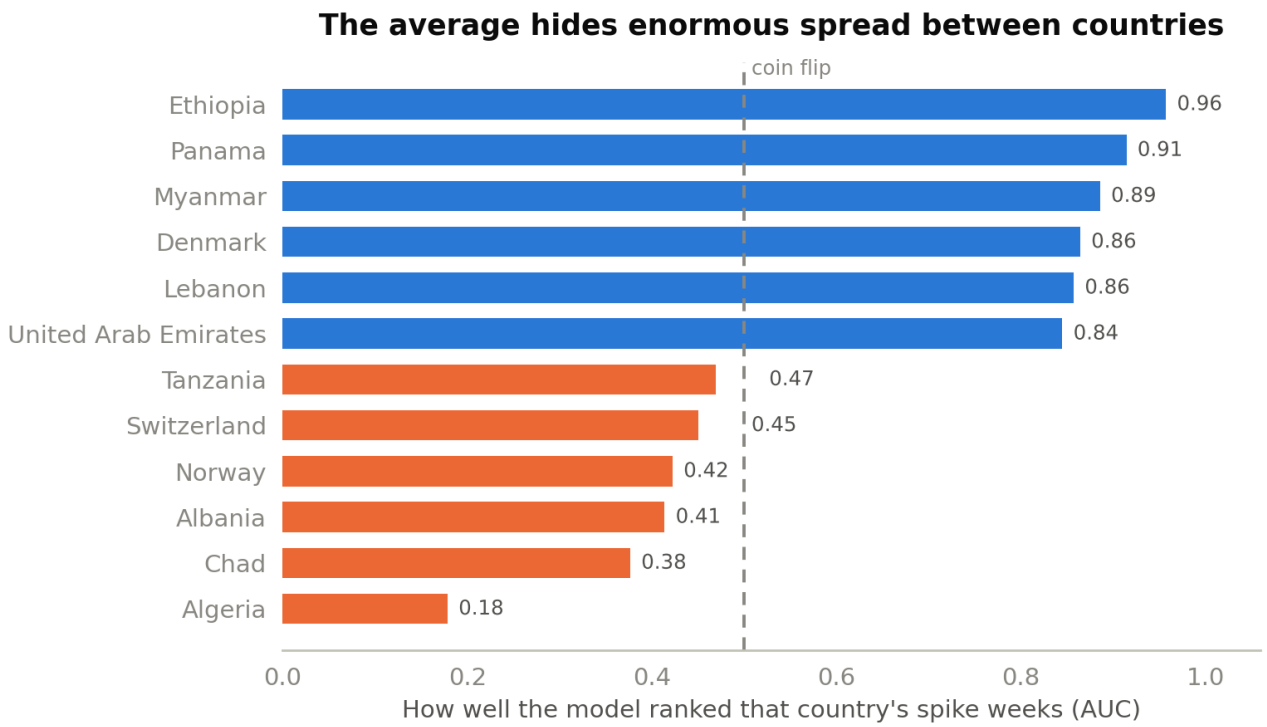
Reordering a watchlist a human will read anyway. One in five beats the one in eleven you get reading countries in arbitrary order, and the scores are calibrated, so you know where to stop.

Not good for

Anything automatic with a cost attached. Four alerts in five are wrong, so this must never trigger a decision on its own. It is a triage aid, not an alarm.

WHERE IT FALLS SHORT

Three things worth stating plainly



It measures coverage, not violence

A spike in coded events can be a spike in attention. No human-coded conflict dataset was used for validation, which is the first thing to fix.

The average describes nobody

Per-country performance runs from 0.96 down to 0.18. 37 of 44 countries beat a coin flip; the rest do not.

One split, ~349 positives

Enough to show the signal exists, not enough for tight confidence intervals. Walk-forward validation is the obvious next step.

Also honest: logistic regression matches the ensemble on average precision. The boosted model's advantage is confined to the high-confidence end, which happens to be the end you operate at.

WHAT COMES NEXT

Ordered by what would settle the questions fastest

1. **Check it against reality**

   Compare the labels with a human-coded conflict dataset such as ACLED or UCDP. This decides how much of the signal is escalation rather than news attention.

2. **Validate over many windows**

   Roll the train/test cutoff forward repeatedly instead of splitting once, and report uncertainty rather than point estimates.

3. **Let it see who, not just where**

   Add actor-level features so the model can watch escalation between specific pairs of states, invisible to it today.

4. **Try a model that reads sequences**

   Give a sequence model the weekly series directly and see whether the ceiling reached here belongs to the data or to the hand-built lags.

The pipeline was built so each of these is an addition rather than a rewrite. That was the point of building it as a pipeline.

THE TAKEAWAY

Free news data really does give you a week or two of warning.

It is a signal about reporting rather than about the world, it is modest in size, and it is worth having anyway, for deciding what to read first rather than for raising alarms.

1.9x
better than guessing

every
horizon beats the rule of thumb

1 in 5
alerts are real

Deven Shah · MET CS 777 · Code, data documentation and the web app are in the project repository